

NIKOLAY BOGOYCHEV

Research Interest: Machine Translation, GPGPU, GEMM, Low/high level performance optimisation

✉ nheart@gmail.com
✉ [@xapajiamnu.bsky.social](https://xapajiamnu.bsky.social)

🇬🇧 United Kingdom
✉ @xapajiamnu@social.linux.pizza

🌐 www.nbogoychev.com
🏠 [Nikolay Bogoychev](#)

🔗 [XapaJlaMnu](#)
🏠 [Nikolay Bogoychev](#)

🔗 [XapaJlaMnu](#)
🏠 [nikolay-bogoychev](#)

📺 [@xapajiamnu](#)

EDUCATION

PhD

School of Informatics, University of Edinburgh

📅 2015 – 2019

📍 Edinburgh, United Kingdom

- Thesis: Fast machine translation on parallel and massively parallel hardware
- Supervisors: Adam Lopez, Kenneth Heafield

BSc

Artificial Intelligence and Computer science, University of Edinburgh

📅 2010 – 2014

📍 Edinburgh, United Kingdom

- First Class Honours

WORK EXPERIENCE

CTO

Last Token

📅 June 2026 - Present

📍 Remote, United Kingdom

Providing premium quality training data for Large language models.

Research Scientist

Meta

📅 December 2023 - December 2025

📍 London, United Kingdom

Working on the LLaMa family of models.

- Released the LLaMa 3.2 1B and 3B models
- Created the [Multiloko](#) multilingual local knowledge evaluation dataset
- Also worked on Pruning, Distillation, Evaluation, Scaling Laws, etc

CTO

Efficient Translation Limited

📅 March 2023 - December 2023

📍 Edinburgh, United Kingdom

Edge device machine translation for Privacy focussed customers.

- x86/arm fast machine translation services deployed on local devices.
- terminology translation.

Postdoctoral Researcher

University of Edinburgh, School of Informatics

📅 January 2019 - December 2023

📍 Edinburgh, United Kingdom

Working on neural machine translation and High performance computing

- [translateLocally](#), cross platform, fast, offline desktop translation software
github.com/XapaJlaMnu/translateLocally
- marian NMT 8bit integer CPU/GPU tensorcore inference code
- CPU int8 VNNI GEMM: github.com/kpu/intgemm
- Experimental GEMM tiling: github.com/XapaJlaMnu/bfTile
- Bergamot: Privacy driven Firefox translation. For details: browser.mt

SKILLS

C++ python CUDA C gdb

x86 assembly java

Machine Translation NLP LLM

Pretraining Evaluation

Deduplication Distillation

Pruning Multilinguality

ACTIVITIES

Open Source Software

I am a firm supporter and believer in Linux and Open source software and I have contributed to a number of open source projects, among which **python**, **wine**, **archlinux**, **marian-nmt**, **moses**.

I am also the author of:

- **TranslateLocally**, a cross-platform fast and secure offline translation.
- **ProbingPT**, a fast phrase table for statistical machine translation
- **gLM** a GPU n -gram language model
- **imageSelector** a cross-platform QT photo library organiser

Privacy driven Translation

I am the principal developer of **TranslateLocally**, a cross-platform fast and secure offline translation client. Give it a try!

<https://nbogoychev.com/bringing-machine-translation-to-the-users/>

Blog

I maintain a blog about Chinese characters, research and coding.

nbogoychev.com

Languages

I am an avid language learner and I am known to speak to a degree and with varying success Bulgarian, English, Spanish, German, Mandarin Chinese, Japanese and Portuguese.

Consultant

Intel

📅 March 2019 - December 2023

📍 Edinburgh, United Kingdom

Exploring opportunities for CPU training and inference for NMT:

- BF16 and other quantisation schemes for training and inference.
- Low level hardware specific optimisation with focus on NMT.

SELECTED PUBLICATIONS

- Dieuwke Hupkes and **Nikolay Bogoychev** (2025). *MultiLoKo: a multilingual local knowledge benchmark for LLMs spanning 31 languages*. arXiv: [2504.10356](#)
- The LLaMa Team (2024). *The Llama 3 Herd of Models*. arXiv: [2407.21783](#)
- **Nikolay Bogoychev**, Pinzhen Chen, et al. (June 2024). "The Ups and Downs of Large Language Model Inference with Vocabulary Trimming by Language Heuristics". In: *Insights from Negative Results in NLP*. ed. by Shabnam Tafreshi, Arjun Akula, et al.
- **Nikolay Bogoychev**, Jelmer van der Linde, et al. (2023). *OpusCleaner and OpusTrainer, open source toolkits for training Machine Translation and Large language models*. arXiv: [2311.14838](#)
- Andreas Grivas, **Nikolay Bogoychev**, and Adam Lopez (2022). "Low-Rank Softmax Can Have Unargmaxable Classes in Theory but Rarely in Practice". In: *ACL*
- **Nikolay Bogoychev** (2021). "Not all parameters are born equal: Attention is mostly what you need". In: *BlackboxNLP*
- **Nikolay Bogoychev**, Jelmer Van der Linde, and Kenneth Heafield (2021). "TranslateLocally: Blazing-fast translation running on the local CPU". in: *EMNLP*
- **Nikolay Bogoychev** and Pinzhen Chen (2021). "The Highs and Lows of Simple Lexical Domain Adaptation Approaches for Neural Machine Translation". In: *Insights from Negative Results in NLP*
- Alham Fikri Aji, **Nikolay Bogoychev**, et al. (July 2020). "In Neural Machine Translation, What Does Transfer Learning Transfer?" In: *ACL*
- **Nikolay Bogoychev**, Roman Grundkiewicz, et al. (July 2020). "Edinburgh's Submissions to the 2020 Machine Translation Efficiency Task". In: *WNGT*
- **Nikolay Bogoychev**, Marcin Junczys-Dowmunt, et al. (Oct. 2018). "Accelerating Asynchronous Stochastic Gradient Descent for Neural Machine Translation". In: *EMNLP*
- **Nikolay Bogoychev** and Adam Lopez (2016). "N-gram language models for massively parallel devices". In: *ACL*

TALKS AND TUTORIALS

The Role of Locally Relevant Content in Evaluating Multilingual LLMs

LocWorld54

📅 09/2025

📍 Monterey, United States

https://nbogoychev.com/files/publications/locworld_multilko.pdf

Efficient Machine translation tutorial

MT Marathon

📅 11/2021

📍 Edinburgh, United Kingdom

nbogoychev.com/efficient-machine-translation/
